Jinyuan Hu 胡锦源

Undergraduate, Computer Sciences, University of Wisconsin–Madison
jhu489@wisc.edu · yelonhu.github.io · github.com/yelonhu · Google Scholar

Research

I work on how a robot policy with a general manipulation prior should use a small amount of new, relevant information—a short demonstration, what it just observed, what its own actions just did—to execute fine, contact-rich manipulation reliably on the first attempt. My background is reinforcement-learning post-training of vision–language models with verifiable rewards and self-generated curricula; my current focus is training recipes (data organization, objectives, memory and action interfaces) that make context carry non-redundant information for action, and separating three explanations of failure that are usually conflated: missing information, information the policy ignores, and limits of action generation and execution timing.

Education

University of Wisconsin–Madison 2026 – May 2028 (expected)
B.S. in Computer Sciences; transfer student.

The Chinese University of Hong Kong 2024 – 2026
Mathematics and Information Engineering; transferred to UW–Madison with 66 credits.

Publications & Preprints

* equal contribution.

- **PhysEvolve: Verifier-Grounded Self-Evolution for Physical Reasoning with Executable Worlds. Jinyuan Hu***, Yilei Jiang*, Chen Tang, Minghong Cai, Xinran Liu, Xiangyu Yue. *Under review, ICLR 2027.* A shared VLM policy proposes executable physical worlds; a deterministic executor runs matched baseline and counterfactual branches and binds exact answers that train the same policy with GRPO (8×A800). Zero-shot transfer from rigid-body training to fluid, cloth and real-world video benchmarks. Led method, training and evaluation; code released.
- **OpenGame: Open Agentic Coding for Games.** Yilei Jiang, **Jinyuan Hu**, Qianyin Xiao, Yaozhi Zheng, Ruize Ma, Kaituo Feng, Jiaming Han, Tianshuo Peng, Kaixuan Fan, Manyuan Zhang, Xiangyu Yue. *Under review, EMNLP 2026; arXiv:2604.18394.* Open-source agentic framework for end-to-end game creation (Game Skill: template scaffolding plus execution-grounded debugging), GameCoder-27B trained with SFT and execution-grounded RL, and OpenGame-Bench. Second author and core contributor; ≈3k GitHub stars.
- **VERSE: Verifiable Spatial Self-Evolution with Geometric Process Rewards.** Xinran Liu, Yilei Jiang, Tianshuo Peng, Kaixuan Fan, **Jinyuan Hu**, Xiangyu Yue. *Under review, ICLR 2027.*

Research Experience

CUHK MMLab, research intern (Prof. Xiangyu Yue) 2025 – present
RL post-training of multimodal models with verifiable rewards; self-evolving training loops (proposer–solver self-play, adaptive curriculum over reasoning intents); large-scale evaluation across PhysBench, ContPhy and CausalPhys. Working remotely from Madison since 2026.

Independent robot-learning work 2026 – present
Built and calibrated an SO-101 leader–follower arm (LeRobot stack); teleoperation data collection and ACT policy training for closed-loop experiments at home. Used as a sanity-check platform for the research question above.

Skills

Learning	RL post-training (GRPO, verifiable rewards, KL-regularized policy optimization), self-play and curriculum design, VLM fine-tuning, imitation learning (ACT, diffusion / flow policies), evaluation-protocol design.
Systems	PyTorch; multi-GPU rollout and training pipelines; sandboxed program execution for verifiers; MuJoCo; LeRobot.
Robotics	SO-101 (Feetech bus servos), teleoperation pipelines, basic kinematics and impedance control.
Languages	Python, C/C++; Mandarin (native), English.

Last updated October 2026.